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EXAMINER	
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/627,662	Applicant(s) AZUMA, Kenta	
	Examiner RACHEL N NGUYEN	Art Unit 3645	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 05 April 2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-19 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-19 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 05 April 2024 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

1. This is the first office action on the merits. Claims 1-19 are currently pending.

Notice of Pre-AIA or AIA Status

2. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Priority

3. Receipt is acknowledged of certified copies of papers required by 37 CFR 1.55.

Information Disclosure Statement

4. The information disclosure statement (IDS) submitted on 4/5/2024 is in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement is being considered by the examiner.

Claim Interpretation

5. The following is a quotation of 35 U.S.C. 112(f):

> (f) Element in Claim for a Combination. – An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

The following is a quotation of pre-AIA 35 U.S.C. 112, sixth paragraph:

> An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

6. The claims in this application are given their broadest reasonable interpretation using the plain meaning of the claim language in light of the specification as it would be understood by one of ordinary skill in the art. The broadest reasonable interpretation of a claim element (also commonly referred to as a claim limitation) is limited by the description in the specification when 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is invoked.

As explained in MPEP § 2181, subsection I, claim limitations that meet the following three-prong test will be interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph:

(A) the claim limitation uses the term "means" or "step" or a term used as a substitute for "means" that is a generic placeholder (also called a nonce term or a non-structural term having no specific structural meaning) for performing the claimed function;

(B) the term "means" or "step" or the generic placeholder is modified by functional language, typically, but not always linked by the transition word "for" (e.g., "means for") or another linking word or phrase, such as "configured to" or "so that"; and

(C) the term "means" or "step" or the generic placeholder is not modified by sufficient structure, material, or acts for performing the claimed function.

Use of the word "means" (or "step") in a claim with functional language creates a rebuttable presumption that the claim limitation is to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites sufficient structure, material, or acts to entirely perform the recited function.

Absence of the word "means" (or "step") in a claim creates a rebuttable presumption that the claim limitation is not to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is not interpreted under 35 U.S.C. 112(f) or pre-

AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites function without reciting sufficient structure, material or acts to entirely perform the recited function.

Claim limitations in this application that use the word "means" (or "step") are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action. Conversely, claim limitations in this application that do not use the word "means" (or "step") are not being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action.

7. This application includes one or more claim limitations that do not use the word "means," but are nonetheless being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, because the claim limitation(s) uses a generic placeholder that is coupled with functional language without reciting sufficient structure to perform the recited function and the generic placeholder is not preceded by a structural modifier. Such claim limitations are:

Level adjustment unit: "Unit" is a generic placeholder as a person of ordinary skill in the art would not know the specific structure for the recited function of "to change an irradiation intensity of the sensing light output from the irradiation unit or detection sensitivity of the photodetectors from a normal level to a suppression level smaller than the normal level by a predetermined amount." Further, "unit" is not modified by sufficient structure. In this case, "unit" is merely modified by the "level adjustment" functional language.

Peak detection unit: "Unit" is a generic placeholder as a person of ordinary skill in the art would not know the specific structure for the recited function of "to detect a received light pulse corresponding to reflected light which is the sensing light reflected by an object and returned, and a peak of the received light pulse, based on time-series data of a number of responses of the photodetector." Further, "unit" is not modified by sufficient structure. In this case, "unit" is merely modified by the "peak detection" functional language.

Pulse information acquisition unit: “Unit” is a generic placeholder as a person of ordinary skill in the art would not know the specific structure for the recited function of “to acquire, as pulse information, a data set indicating a predetermined feature amount related to the received light pulse detected by the peak detection unit, the pulse information including normal pulse information at the normal level and suppression pulse information at the suppression level.” Further, “unit” is not modified by sufficient structure. In this case, “unit” is merely modified by the “pulse information acquisition” functional language.

Distance calculation unit: “Unit” is a generic placeholder as a person of ordinary skill in the art would not know the specific structure for the recited function of “to calculate a distance value to the target based on the normal pulse information and the suppression pulse information.” Further, “unit” is not modified by sufficient structure. In this case, “unit” is merely modified by the “distance calculation” functional language.

Because these claim limitations are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, they are being interpreted to cover the corresponding structure described in the specification as performing the claimed function, and equivalents thereof.

The means-plus limitations are interpreted as follow:

Level adjustment unit: Level adjustment unit F3 (Fig. 9 and Paragraph [0083])

Peak detection unit: peak detector 8 (Fig. 1 and Paragraph [0056], with further details in Figs. 4-5 and corresponding paragraphs)

Pulse information acquisition unit: Pulse information acquisition unit F2 (Fig. 9 and Paragraph [0082])

Distance calculation unit: Distance calculation unit F4 (Fig. 9 and Paragraph [0085])

If applicant does not intend to have these limitations interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, applicant may: (1) amend the claim limitation(s) to avoid it/them

being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph (e.g., by reciting sufficient structure to perform the claimed function); or (2) present a sufficient showing that the claim limitation(s) recite(s) sufficient structure to perform the claimed function so as to avoid it/them being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph.

Claim Rejections - 35 USC § 103

8. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

9. The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

10. Claims 1-4, 11-13, 16, and 19 are rejected under 35 U.S.C. 103 as being unpatentable over Sumi et al., JP 2015194356 A (“Sumi”) in view of Rosenzweig et al., US 20210025997 A1 (“Rosenzweig”).

Regarding claim 1, Sumi discloses A light distance measurement device configured to detect a distance to a target by using a round-trip time of light to the target (Fig. 1, distance measuring device, Paragraph [0013]), the light distance measurement device comprising:

an irradiation unit configured to irradiate sensing light, which is light having a predetermined wavelength, in a predetermined detection target direction (Fig. 1, light emitting unit 200, Paragraph [0017]);

photodetectors ~~arranged in a matrix~~ and configured to respond to the sensing light (Fig. 1, light receiving element 301, Paragraph [0018]);

[...];

a peak detection unit configured to detect a received light pulse corresponding to reflected light which is the sensing light reflected by an object and returned, and a peak of the received light pulse, based on time-series data of a number of responses of the photodetector (Fig. 1, detecting unit 400, Paragraph [0019]);

a pulse information acquisition unit configured to acquire, as pulse information, a data set indicating a predetermined feature amount related to the received light pulse detected by the peak detection unit, the pulse information including normal pulse information at the normal level and suppression pulse information at the suppression level (Fig. 1, detecting unit 400, Paragraph [0019]); and

a distance calculation unit configured to calculate a distance value to the target based on the normal pulse information and the suppression pulse information (Fig. 1, time measurement unit 500, Paragraph [0020]).

Sumi does not teach: photodetectors arranged in a matrix and a level adjustment unit configured to change an irradiation intensity of the sensing light output from the irradiation unit or

detection sensitivity of the photodetectors from a normal level to a suppression level smaller than the normal level by a predetermined amount.

However, Rosenzweig teaches a LIDAR system with a sensor that contains photodetectors arranged in a matrix that detects both external and internal light reflections (Fig. 7B, sensor 712, detectors 708, Paragraph [0147]; See also: Paragraph [0099]). In determining that performance has degraded due to internal reflections, the processor changes the irradiation intensity of the sensing light output from the irradiation unit or changes the detection sensitivity of the photodetectors from a normal level to a suppression level smaller than the normal level by a predetermined amount (Fig. 9, step 911, Paragraph [0186]-[0187]).

It would have been obvious to someone of ordinary skill in the art before the effective filing date of the claimed invention to have modified Sumi's distance measuring device by adding an array of photodetectors and a processing unit which changes the light intensity or photodetector sensitivity in response to performance degradation, which is disclosed by Rosenzweig. One of ordinary skill in the art would have been motivated to make this modification in order to increase the signal-to-noise ratio, as suggested by Rosenzweig (Paragraph [0191]).

Regarding claim 2, Sumi, as modified in view of Rosenzweig, discloses The light distance measurement device according to claim 1, wherein

the normal pulse information includes a pulse width (Sumi, Fig. 1, time measurement unit 500, signal processing unit 100, Paragraph [0020]),

the distance calculation unit is configured to:

determine whether a target pulse, which is the received light pulse to be processed, is a combination of the reflected light from the target and unwanted reflected light based on the pulse

width included in the normal pulse information for the target pulse (Sumi, Fig. 3, first comparator 402, second comparator 403, Paragraph [0022]; Fig. 5, S504, Paragraph [0030]); and

change at least one of a calculation formula and a feature amount for calculating the distance value according to whether the target pulse is a combination of the reflected light from the target and the unnecessary reflected light (Sumi, Fig. 5, S506 to S503, Paragraph [0030], [0020]; See also Eq. 1-3, Paragraph [0003], [0028]-[0029]).

Regarding claim 3, Sumi, as modified in view of Rosenzweig, discloses The light distance measurement device according to claim 2, wherein

the distance calculation unit is configured to determine that the target pulse is the combination of the reflected light from the target and the unnecessary reflected light based on the pulse width included in the normal pulse information for the target pulse being equal to or greater than a predetermined value (Sumi, Figs. 5-6, S504, Paragraph [0030]; See also: Equations 2-3).

Regarding claim 4, Sumi, as modified in view of Rosenzweig, discloses The light distance measurement device according to claim 2, wherein

the suppression pulse information includes the pulse width (Sumi, Fig. 3, Equation 3, Paragraph [0027], [0029]), and

the distance calculation unit is configured to determine that the target pulse is the combination of the reflected light from the target and the unnecessary reflected light, based on a difference between the pulse width included in the normal pulse information and the pulse width included in the suppression pulse information for the target pulse being equal to or greater than a predetermined value (Sumi, Figs. 5-6, S504 to S506, Paragraph [0030]).

Regarding claim 11, Sumi, as modified in view of Rosenzweig, discloses The light distance measurement device according to claim 2, wherein

the normal pulse information includes a rise determination time indicating a timing at which the intensity of the received light pulse is equal to or greater than the threshold value (Sumi, Fig. 1, time measurement unit 500, High threshold time measurement unit 501, Low threshold time measurement unit 502, Paragraph [0020]; Fig. 3, Paragraph [0022]), and

the distance calculation unit is configured to calculate the distance value using the rise determination time included the normal pulse information when it is determined that the target pulse is the reflected light from the target to which the unnecessary reflected light is not combined (Sumi, Fig. 5, S504 to S505, Paragraph [0030]).

Regarding claim 12, Sumi, as modified in view of Rosenzweig, discloses The light distance measurement device according to claim 11, wherein

the distance calculation unit is configured to determine the distance value by correcting a value, which is obtained by multiplying the rise determination time by half of the speed of light (Fig. 1, time measurement unit 500, signal processing unit 100, Paragraph [0020], Equation 1, Paragraph [0003]), according to the pulse width when it is determined that the target pulse is the reflected light from the target to which the unnecessary reflected light is not combined (Fig. 5, S504, Paragraph [0030]), and

the distance calculation unit is configured not to perform the correction using the pulse width when it is determined that the target pulse is the combination of the reflected light from the target and the unnecessary reflected light (Fig. 5, S505, Paragraph [0030]).

Regarding claim 13, Sumi, as modified in view of Rosenzweig, discloses The light distance measurement device according to claim 1, wherein

the normal pulse information includes peak intensity indicating intensity at a peak (Sumi, Paragraph [0027]), and

the distance calculation unit is configured to:

calculate the distance value using the suppression pulse information when the peak intensity included in the normal pulse information is equal to or greater than a predetermined value (Sumi, Fig. 5, S501 to S502, Paragraph [0030]); and

calculate the distance value using the normal pulse information when the peak intensity included in the normal pulse information is less than the predetermined value (Sumi, Fig. 5, S501 to S505, Paragraph [0030]).

Regarding claim 16, Sumi, as modified in view of Rosenzweig, discloses The light distance measurement device according to claim 1.

Sumi, as modified in view of Rosenzweig, does not teach: the distance calculation unit is configured to determine that the received light pulse is the unnecessary reflected light when the received light pulse observed within a predetermined time after the sensing light is irradiated at the normal level is not observed when the suppression level is applied.

However, Rosenzweig teaches a LIDAR system that determines that performance has degraded due to internal reflections, the processor changes the irradiation intensity of the sensing light output from the irradiation unit or changes the detection sensitivity of the photodetectors from a normal level to a suppression level smaller than the normal level by a predetermined amount (Fig. 9, step 911, Paragraph [0186]-[0187]). The signal-to-noise is increased when the irradiation intensity is increased in response to sensor degradation from unnecessary reflected light (Paragraph [0191]).

It would have been obvious to someone of ordinary skill in the art before the effective filing date of the claimed invention to have modified Sumi's distance measuring device by adding an array of

photodetectors and a processing unit which changes the light intensity or photodetector sensitivity in response to performance degradation, which is disclosed by Rosenzweig. One of ordinary skill in the art would have been motivated to make this modification in order to increase the signal-to-noise ratio, as suggested by Rosenzweig (Paragraph [0191]).

Regarding claim 19, Sumi, as modified in view of Rosenzweig, discloses The light distance measurement device according to claim 1, wherein the distance calculation unit is configured to:

determine whether multiple reflected light is likely to be received by comparing the normal pulse information and the suppression pulse information (Sumi, Fig. 5, S501, S504, Paragraph [0030]); and

perform a process of notifying the occupant, the operator present outside the vehicle, or another device that accuracy of distance measurement to the target is reduced when it is determined that the multiple reflected light is likely to be received (Sumi, Paragraph [0024]).

11. Claims 5-10 and 17-18 are rejected under 35 U.S.C. 103 as being unpatentable over Sumi in view of Rosenzweig in further view of Nomura, US 20190178996 A1 (“Nomura”).

Regarding claim 5, Sumi, as modified in view of Rosenzweig, discloses The light distance measurement device according to claim 2, wherein

the distance calculation unit is configured to:

[...]; and

change at least one of the calculation formula and the feature amount used for calculating the distance value according to the determined type of the unnecessary reflected light (Sumi, Figs. 5-6, S506 to S503, Paragraph [0030]).

Sumi, as modified in view of Rosenzweig, does not teach: determine a type of the unnecessary reflected light by comparing the normal pulse information and the suppression pulse information for the target pulse when it is determined that the target pulse is the combination of the reflected light from the target and the unnecessary reflected light.

However, Nomura teaches a distance measuring device that uses two comparators to determine the rise and fall times of a received signal containing normal pulse information and suppression pulse information (Figs. 3A and 4A, signal s1, signal s2, signal s3, noise n1, Paragraph [0037]). Nomura teaches comparing the calculated distance from the received pulse to distance thresholds to determine if the type of unnecessary reflected light is reflected from a dirty window or other noise (Fig. 5, P10, P11, P12, p14, p16, Paragraph [0052]-[0054]).

It would have been obvious to someone of ordinary skill in the art before the effective filing date of the claimed invention to have modified the distance measuring method disclosed by Sumi and Rosenzweig by adding steps to distinguish the type of unnecessary reflected light, which is disclosed by Nomura. One of ordinary skill in the art would have been motivated to make this modification in order to notify the user to the possibility of dirt on the window material, as suggested by Nomura (Paragraph [0052]).

Regarding claim 6, Sumi, as modified in view of Rosenzweig and Nomura, discloses The light distance measurement device according to claim 5, wherein

each of the normal pulse information and the suppression pulse information includes a peak arrival time which is a time from irradiation of the sensing light to observation of a peak (Sumi, Paragraph [0027]), and

the distance calculation unit is configured to determine the type of the unnecessary reflected light by comparing the peak arrival time included in the normal pulse information for the target pulse

and the peak arrival time included in the suppression pulse information when it is determined that the target pulse is the combination of the reflected light from the target and the unnecessary reflected light (Nomura, Fig. 5, P10, P11, P12, p14, p16, Paragraph [0052]-[0054]; See also Paragraph [0050]).

Regarding claim 7, Sumi, as modified in view of Rosenzweig and Nomura, discloses The light distance measurement device according to claim 6, wherein

each of the normal pulse information and the suppression pulse information further includes a rise determination time indicating a timing at which intensity of the received light pulse is equal to or greater than a threshold value and a fall determination time indicating a timing at which the intensity of the received light pulse is equal to or lower than the threshold value (Sumi, Fig. 1, time measurement unit 500, High threshold time measurement unit 501, Low threshold time measurement unit 502, Paragraph [0020]), and

the distance calculation unit is configured to:

determine whether the unnecessary reflected light is proximity body scattered light, which is scattered light caused by an adhered matter on an irradiation window or scattered light inside a housing, by comparing the peak arrival time included in the normal pulse information for the target pulse and the peak arrival time included in the suppression pulse information (Nomura, Fig. 5, P10, P11, P12, p14, p16, Paragraph [0052]-[0054]; See also Paragraph [0050]);

calculate the distance value using the fall determination time when it is determined that the unnecessary reflected light is the proximity body scattered light (Sumi, Fig. 8, S804 to S806, Paragraph [0032]); and

calculate the distance value using the rise determination time when it is determined that the unnecessary reflected light is not the proximity body scattered light (Sumi, Fig. 5, S504 to S506, Paragraph [0030]).

Regarding claim 8, Sumi, as modified in view of Rosenzweig and Nomura, discloses The light distance measurement device according to claim 5, wherein

each of the normal pulse information and the suppression pulse information includes a rise determination time indicating a timing at which the intensity of the received light pulse is equal to or greater than the threshold value (Sumi, Fig. 1, time measurement unit 500, High threshold time measurement unit 501, Low threshold time measurement unit 502, Paragraph [0020]), and

the distance calculation unit is configured to determine the type of the unnecessary reflected light by comparing the rise determination time included in the normal pulse information and the rise determination time included in the suppression pulse information for the target pulse when it is determined that the target pulse is the combination of the reflected light from the target and the unnecessary reflected light (Fig. 5, S501, Paragraph [0030]).

Regarding claim 9, Sumi, as modified in view of Rosenzweig and Nomura, discloses The light distance measurement device according to claim 8, wherein

each of the normal pulse information and the suppression pulse information further includes a fall determination time indicating a timing at which the intensity of the received light pulse is equal to or less than the threshold value (Sumi, Fig. 1, time measurement unit 500, High threshold time measurement unit 501, Low threshold time measurement unit 502, Paragraph [0020]), and

the distance calculation unit is configured to:

determine whether the unnecessary reflected light is proximity body scattered light, which is scattered light caused by an adhered matter on an irradiation window or scattered light inside a housing, by comparing the rise determination time included in the normal pulse information for the

target pulse and the fall determination time included in the suppression pulse information (Nomura, Fig. 5, P10, P11, P12, p14, p16, Paragraph [0052]-[0054]; See also Paragraph [0050]);

calculate the distance value using the fall determination time when it is determined that the unnecessary reflected light is the proximity body scattered light (Sumi, Fig. 8, S804 to S806, Paragraph [0032]); and

calculate the distance value using the rise determination time when it is determined that the unnecessary reflected light is not the proximity body scattered light (Sumi, Fig. 5, S504 to S506, Paragraph [0030]).

Regarding claim 10, Sumi, as modified in view of Rosenzweig, discloses The light distance measurement device according to claim 2, wherein

[...]; and

change at least one of the calculation formula and the feature amount for calculating the distance value according to whether the unnecessary reflected light is the proximity body scattered light (Sumi, Figs. 5-6, S506 to S503, Paragraph [0030]).

Sumi, as modified in view of Rosenzweig, does not teach: the distance calculation unit is configured to: determine whether the unnecessary reflected light is proximity body scattered light, which is scattered light caused by an adhered matter on an irradiation window or scattered light inside a housing, by comparing the normal pulse information for the target pulse and the suppression pulse information when it is determined that the target pulse is the combination of the reflected light from the target and the unnecessary reflected light.

However, Nomura teaches a distance measuring device that uses two comparators to determine the rise and fall times of a received signal containing normal pulse information and suppression pulse information (Figs. 3A and 4A, signal s1, signal s2, signal s3, noise n1, Paragraph [0037]). Nomura teaches

comparing the calculated distance from the received pulse to distance thresholds to determine if the type of unnecessary reflected light is reflected from a dirty window or other noise (Fig. 5, P10, P11, P12, p14, p16, Paragraph [0052]-[0054]).

It would have been obvious to someone of ordinary skill in the art before the effective filing date of the claimed invention to have modified the distance measuring method disclosed by Sumi and Rosenzweig by adding steps to distinguish the type of unnecessary reflected light, which is disclosed by Nomura. One of ordinary skill in the art would have been motivated to make this modification in order to notify the user to the possibility of dirt on the window material, as suggested by Nomura (Paragraph [0052]).

Regarding claim 17, Sumi, as modified in view of Rosenzweig, discloses The light distance measurement device according to claim 1.

Sumi, as modified in view of Rosenzweig, does not teach: the distance calculation unit is configured to:

detect the adhered matter on the irradiation window by comparing the normal pulse information and the suppression pulse information; and

perform a process for cleaning the irradiation window when the adhered matter is detected.

However, Nomura teaches a distance measuring device that uses two comparators to determine the rise and fall times of a received signal containing normal pulse information and suppression pulse information (Figs. 3A and 4A, signal s1, signal s2, signal s3, noise n1, Paragraph [0037]). Nomura teaches comparing the calculated distance from the received pulse to distance thresholds to determine if the type of unnecessary reflected light is reflected from adhered matter on the irradiation window and perform a process for cleaning the irradiation window by notifying that the window is dirty (Fig. 5, P10, P11, P12, p14, p16, Paragraph [0052]-[0054]).

It would have been obvious to someone of ordinary skill in the art before the effective filing date of the claimed invention to have modified the distance measuring method disclosed by Sumi and Rosenzweig by adding steps to distinguish the type of unnecessary reflected light, which is disclosed by Nomura. One of ordinary skill in the art would have been motivated to make this modification in order to notify the user to the possibility of dirt on the window material, as suggested by Nomura (Paragraph [0052]).

Regarding claim 18, Sumi, as modified in view of Rosenzweig, discloses The light distance measurement device according to claim 1.

Sumi, as modified in view of Rosenzweig, does not teach: the distance calculation unit is configured to:

detect the adhered matter on the irradiation window by comparing the normal pulse information and the suppression pulse information; and

perform a process of notifying an occupant, an operator present outside a vehicle, or another device that the adhered matter is present when the adhered matter is detected.

However, Nomura teaches a distance measuring device that uses two comparators to determine the rise and fall times of a received signal containing normal pulse information and suppression pulse information (Figs. 3A and 4A, signal s1, signal s2, signal s3, noise n1, Paragraph [0037]). Nomura teaches comparing the calculated distance from the received pulse to distance thresholds to determine if the type of unnecessary reflected light is reflected from adhered matter on the irradiation window and perform a process for cleaning the irradiation window by notifying that the window is dirty (Fig. 5, P10, P11, P12, p14, p16, Paragraph [0052]-[0054]).

It would have been obvious to someone of ordinary skill in the art before the effective filing date of the claimed invention to have modified the distance measuring method disclosed by Sumi and

Rosenzweig by adding steps to distinguish the type of unnecessary reflected light, which is disclosed by Nomura. One of ordinary skill in the art would have been motivated to make this modification in order to notify the user to the possibility of dirt on the window material, as suggested by Nomura (Paragraph [0052]).

12. Claims 14-15 are rejected under 35 U.S.C. 103 as being unpatentable over Sumi in view of Rosenzweig in further view of Sumi et al., JP 2015200555 A ("Sumi '555").

Regarding claim 14, Sumi, as modified in view of Rosenzweig, discloses The light distance measurement device according to claim 1, wherein

each of the normal pulse information and the suppression pulse information includes a rise determination time indicating a timing at which the intensity of the received light pulse is equal to or greater than the threshold value (Sumi, Fig. 1, time measurement unit 500, High threshold time measurement unit 501, Low threshold time measurement unit 502, Paragraph [0020]; Fig. 3, Paragraph [0022]), and

[...].

Sumi, as modified in view of Rosenzweig, does not teach: the distance calculation unit is configured to change at least one of the calculation formula and the feature amount for calculating the distance value according to whether a rise time difference is less than a predetermined value, the rise time difference being a difference between the rise determination time included in the normal pulse information and the rise determination time included in the suppression pulse information for a target pulse, which is the received light pulse to be processed.

However Sumi '555 teaches a distance metrology device that decomposes a wave consisting of a normal pulse and suppression pulse. Sumi '555 teaches a method to determine if a normal pulse and

suppression pulse are overlapped by determining if the difference between a first zero-crossing and a second measurement time is less than a predetermined value (Fig. 8, S801, Paragraph [0044]). While Sumi '555 uses the difference between the zero-crossing and a measurement time, one of ordinary skill in the art using Sumi's distance measuring device would recognize that taking the difference between two rise determination times would equivalently indicate the degree of overlap between two signal pulses.

It would have been obvious to someone of ordinary skill in the art before the effective filing date of the claimed invention to have modified the distance measuring method disclosed by Sumi and Rosenzweig by adding steps to change the distance calculation formula based on the degree of overlap between two pulses being less than a predetermined amount, which is disclosed by Sumi '555. One of ordinary skill in the art would have been motivated to make this modification in order to separate the desired signal from the clutter, as suggested by Sumi '555 (Paragraph [0047]).

Regarding claim 15, Sumi, as modified in view of Rosenzweig and Sumi '555, discloses The light distance measurement device according to claim 14, wherein

each of the normal pulse information and the suppression pulse information further includes at least one of a fall determination time indicating a timing at which the intensity of the received light pulse is equal to or less than the threshold value and a peak arrival time which is a time from irradiation of the sensing light to observation of a peak (Sumi, Fig. 1, time measurement unit 500, High threshold time measurement unit 501, Low threshold time measurement unit 502, Paragraph [0020]), and

the distance calculation unit is configured to:

calculate the distance value using the rise determination time when the rise time difference is less than a predetermined value (Sumi, Fig. 5, S504 to S505, Paragraph [0030]); and

calculate the distance value using a parameter other than the rise determination time when the rise time difference is equal to or greater than the predetermined value (Sumi, Fig. 5, S504 to S506, Paragraph [0030]).

Conclusion

13. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

- Sagawa et al., JP 2004184333 A discloses correcting a distance measurement of a received pulse by adding a correction time into the distance calculation.

14. Any inquiry concerning this communication or earlier communications from the examiner should be directed to RACHEL N NGUYEN whose telephone number is (571)270-5405. The examiner can normally be reached Monday - Friday 8 am - 5:30 pm ET.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Yuqing Xiao can be reached at (571) 270-3603. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or

571-272-1000.

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